

Learning Representations from Noisy Data and Brain Imaging: Subspace Modeling for Heteroscedastic Data & Deep Learning for Functional MRI in Alzheimer's Disease

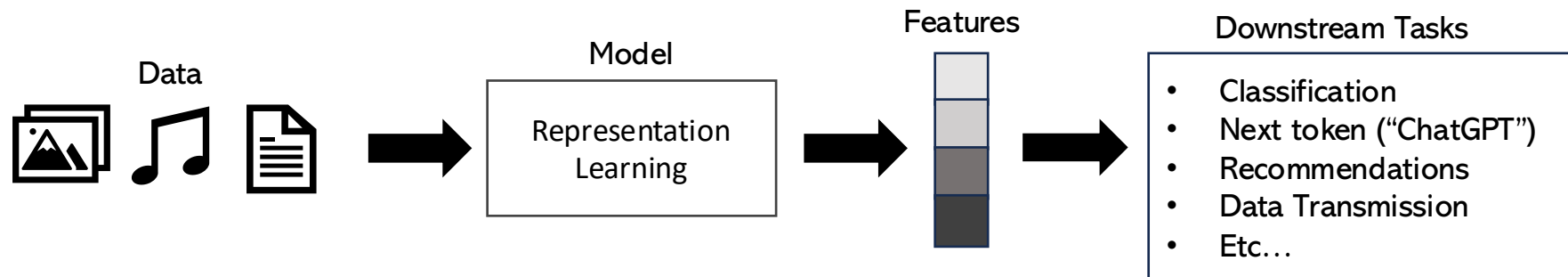
Javier Salazar Cavazos

Department of Electrical & Computer Engineering (ECE)

University of Michigan

March 16th, 2026

Representation Learning



- Part 1 covers explicit modeling (subspace learning)
- Part 2 covers implicit modeling (deep learning)
- Both are flavors of representation learning

Dissertation

Part 1

Chapter 3: ALPCA-H: Subspace Learning for Heteroscedastic Data^{1,2}

Chapter 4: ALPCA-HUS: Subspace Clustering for Heteroscedastic Data

Chapter 5: PET-TURTLE: Deep Unsupervised Support Vector Machines for Imbalanced Data Clusters

Chapter 6: Alzheimer's Disease Classification in Functional MRI With 4D Joint Temporal-Spatial Kernels in Novel 4D CNN Model

Chapter 7: Behavior Score Prediction in Resting-State Functional MRI by Deep State Space Modeling

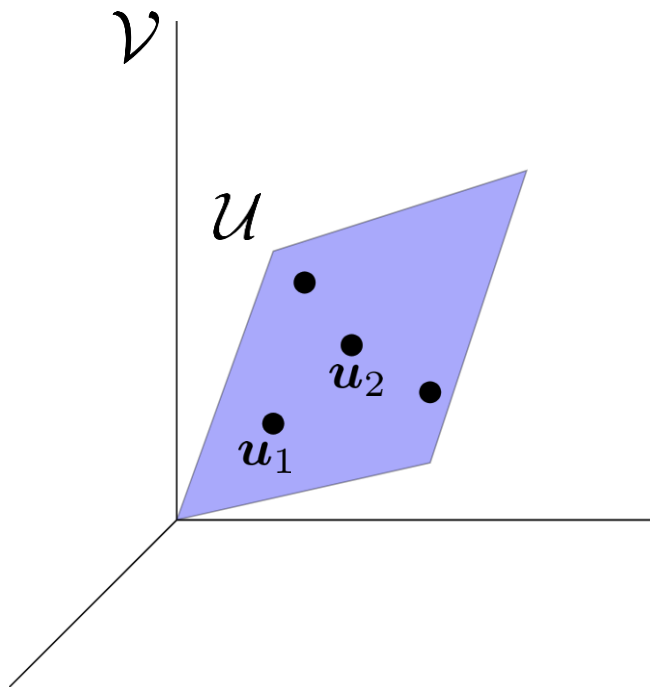
[1] J. Salazar Cavazos, J. A. Fessler and L. Balzano, "ALPCA-H: Sample-wise Heteroscedastic PCA with Tail Singular Value Regularization," [2023 International Conference on Sampling Theory and Applications \(SampTA\)](#), doi: 10.1109/SampTA59647.2023.10301206

[2] J. Salazar Cavazos, J. A. Fessler and L. Balzano, "ALPCA-H: Subspace Learning for Sample-Wise Heteroscedastic Data," in [IEEE Transactions on Signal Processing](#), vol. 73, pp. 876-886, 2025, doi: 10.1109/TSP.2025.3537867

Background

Subspace Learning & Heteroscedastic Data

Subspace Learning



A 2D subspace in a 3D ambient space.

Motivation

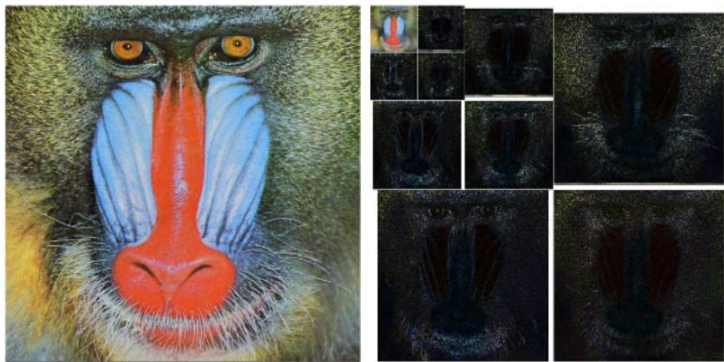
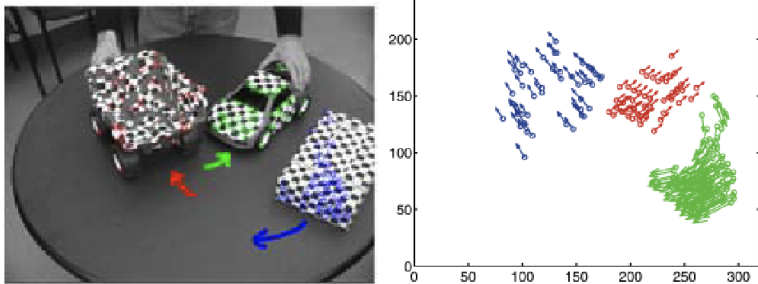


Image Compression¹



Motion Estimation²



Netflix Prize³

[1] Hong, Wei, et al. "Multiscale hybrid linear models for lossy image representation." *IEEE Transactions on Image Processing* 15.12 (2006): 3655-3671.

[2] Vidal, René, Roberto Tron, and Richard Hartley. "Multiframe motion segmentation with missing data using PowerFactorization and GPCA." *International Journal of Computer Vision* 79.1 (2008): 85-105.

[3] Koren, Yehuda, Robert Bell, and Chris Volinsky. "Matrix factorization techniques for recommender systems." *Computer* 42.8 (2009): 30-37.

Heteroscedasticity

y : measurement (known)

x : true value (unknown)

ϵ : noise (unknown)

$i \in \{1, \dots, N\}$: index for data sample

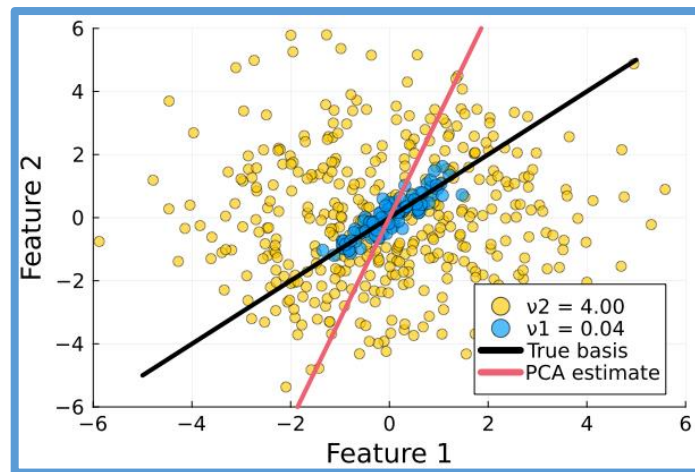
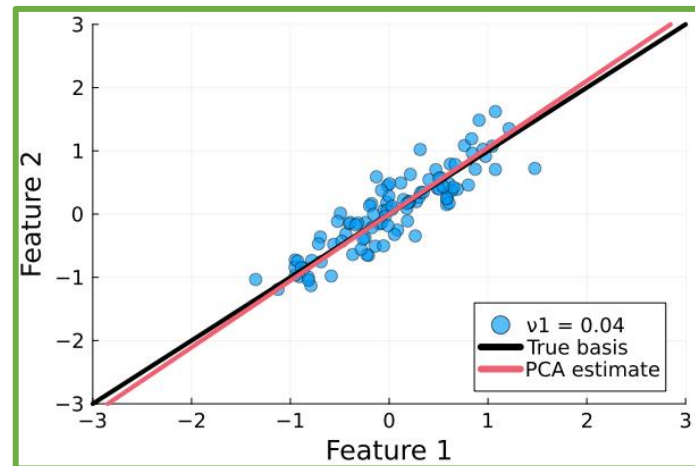
Homoscedastic Data

$$y_i = x_i + \epsilon \quad \text{where} \quad \epsilon \sim \overbrace{\mathcal{N}(0, \nu \mathbf{I})}^{\text{"bell curve"}}$$

Heteroscedastic* Data

$$y_i = x_i + \epsilon_i \quad \text{where} \quad \epsilon_i \sim \mathcal{N}(0, \nu_i \mathbf{I})$$

*other forms of heteroscedasticity exist



Subspace Quality Impact

U : subspace basis

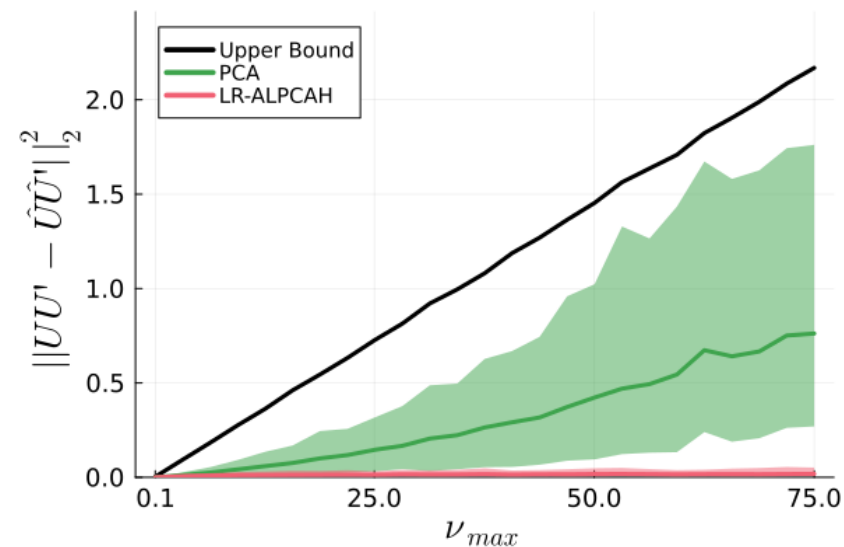
ν : noise variances

D : ambient dimension

$\sigma_d(\cdot)$: d-th singular value

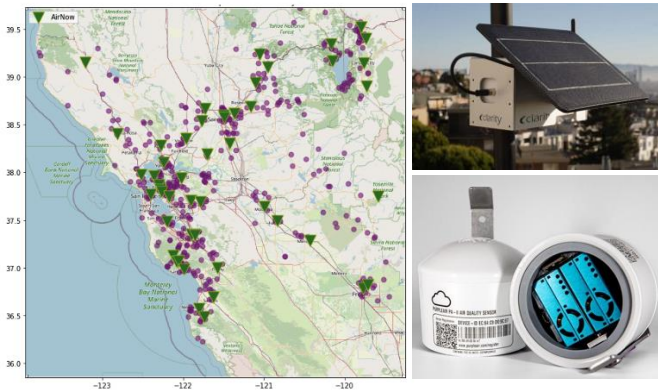
C_X : covariance matrix of $\frac{1}{N} \mathbf{X} \mathbf{X}'$

$$\mathbb{E}[\|\underbrace{U_{:,1:d} U'_{:,1:d}}_{\text{true subspace}} - \underbrace{\hat{U}_{:,1:d} \hat{U}'_{:,1:d}}_{\text{PCA estimate}}\|_2^2] \lesssim \frac{\left(\sqrt{\nu_{\text{sum}}^{(1)}} + \sqrt{D \nu_{\text{max}}} + \sqrt[4]{3D \nu_{\text{sum}}^{(2)}} \right)^2}{N \sigma_d(C_X)}$$



Heteroscedastic Applications

Environmental Air Pollution¹



Lucky imaging²



with



without

X-Ray Applications³

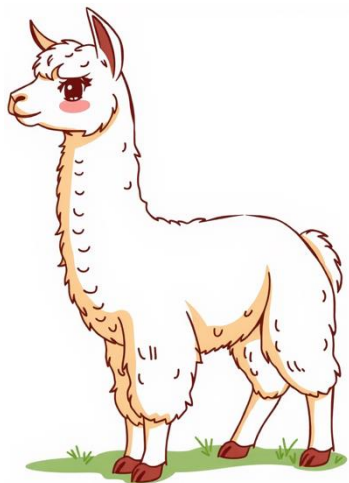
Low-dose CT



High-dose CT



- [1] Hong, David, et al. "HePPCAT: Probabilistic PCA for data with heteroscedastic noise." *IEEE Transactions on Signal Processing* 69 (2021): 4819-4834.
- [2] https://en.wikipedia.org/wiki/Lucky_imaging
- [3] <https://ictprojects.mt/abstracts/2021/a-model-to-improve-low-dose-ct-scan-images/>



Method
ALPCA
(Algorithm for Low-rank PCA for Heteroscedastic Data)

FLUX.1 generated image with prompt “an illustration of an alpaca”

Maximum Likelihood (MLE)

Given the measurement model:

$$\mathbf{y}_i | \mathbf{x}_i \sim \mathcal{N}(\mathbf{x}_i, \boldsymbol{\nu}_i \mathbf{I})$$

$\mathbf{y}$: measurement (known)

$\mathbf{x}$: true value (unknown)

$\boldsymbol{\nu}$: noise variances (unknown)

The probability density function for a single point is:

$$\frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\nu}_i \mathbf{I}|}} \exp \left[-\frac{1}{2} (\mathbf{y}_i - \mathbf{x}_i)' (\boldsymbol{\nu}_i \mathbf{I})^{-1} (\mathbf{y}_i - \mathbf{x}_i) \right]$$

For all data*, after dropping constants, the log-likelihood is:

$$\sum_{i=1}^N -\frac{1}{2} \log |\boldsymbol{\nu}_i \mathbf{I}| - \frac{1}{2} (\mathbf{y}_i - \mathbf{x}_i)' (\boldsymbol{\nu}_i \mathbf{I})^{-1} (\mathbf{y}_i - \mathbf{x}_i)$$

*independence assumption

Maximum Likelihood (MLE)

Assembling things into matrix form:

Let $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_N] \in \mathbb{R}^{D \times N}$

Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N] \in \mathbb{R}^{D \times N}$

Let $\mathbf{\Pi} = \text{diag}(\nu_1, \dots, \nu_N) \in \mathbb{R}^{N \times N}$

$\mathbf{y}$: measurement (known)

$\mathbf{x}$: true value (unknown)

ν : noise variances (unknown)

Tr : matrix trace

$|\cdot|$: matrix determinant

The negative log-likelihood, using matrix properties, is rewritten as:

$$\sum_{i=1}^N -\frac{1}{2} \log |\nu_i \mathbf{I}| - \frac{1}{2} (\mathbf{y}_i - \mathbf{x}_i)' (\nu_i \mathbf{I})^{-1} (\mathbf{y}_i - \mathbf{x}_i) \rightarrow \frac{D}{2} \log |\mathbf{\Pi}| + \frac{1}{2} \text{Tr}[(\mathbf{Y} - \mathbf{X}) \mathbf{\Pi}^{-1} (\mathbf{Y} - \mathbf{X})']$$

By trace lemmas, this can be simplified to:

$$\frac{D}{2} \log |\mathbf{\Pi}| + \frac{1}{2} \|(\mathbf{Y} - \mathbf{X}) \mathbf{\Pi}^{-1/2}\|_F^2$$

Maximum A Posteriori (MAP)

$\mathbf{Y}$: measurement (known)

$\mathbf{X}$: low-rank structure (unknown)

$\mathbf{\Pi}$: noise variances (unknown)

D : ambient dimension

N : sample space

So far, we have not imposed any conditions on $\mathbf{X}$:

$$\frac{D}{2} \log |\mathbf{\Pi}| + \frac{1}{2} \|(\mathbf{Y} - \mathbf{X})\mathbf{\Pi}^{-1/2}\|_F^2$$

We assume the data is low rank, and encourage this prior belief by regularization:

$$\|\mathbf{X}\|_*$$

Nuclear Norm

or

$$\mathbf{X} \approx \mathbf{L}\mathbf{R}' \quad \text{where} \quad \mathbf{L} \in \mathbb{R}^{D \times \hat{d}}, \mathbf{R} \in \mathbb{R}^{N \times \hat{d}}$$

Matrix Factorization

$$\hat{d} \ll \min(D, N)$$

Thus, the optimization problem we propose is the following:

$$\min_{\mathbf{L}, \mathbf{R}, \mathbf{\Pi}} \frac{1}{2} \|(\mathbf{Y} - \mathbf{L}\mathbf{R}')\mathbf{\Pi}^{-1/2}\|_F^2 + \frac{D}{2} \log |\mathbf{\Pi}|$$

Goals: Find $\{\mathbf{L}, \mathbf{R}\}$ matrices for various uses:

1. $\mathbf{L}\mathbf{R}' \approx \mathbf{X}$ is a denoised $\mathbf{Y}$
2. $\text{GramSchmidt}(\mathbf{L}) = \mathbf{U}$
3. Compute basis coefficients $\mathbf{U}'\mathbf{Y}$

ALPCA^H*

$\mathbf{Y}$: measurement (known)

$\mathbf{X}$: low-rank structure (unknown)

$\mathbf{\Pi}$: noise variances (unknown)

Our algorithm for solving cost function below is named **ALPCA^H** (Algorithm for Low-rank PCA for Heteroscedastic data)

$$\min_{\mathbf{L}, \mathbf{R}, \mathbf{\Pi}} \underbrace{\frac{1}{2} \|(\mathbf{Y} - \mathbf{L}\mathbf{R}')\mathbf{\Pi}^{-1/2}\|_F^2 + \frac{D}{2} \log |\mathbf{\Pi}|}_{f(\mathbf{L}, \mathbf{R}, \mathbf{\Pi})}$$

We solve by alternating minimization of the three blocks:

$$\mathbf{L}_{t+1} = \arg \min_{\mathbf{L}} f(\mathbf{L}, \mathbf{R}_t, \mathbf{\Pi}_t) = \mathbf{Y}\mathbf{\Pi}_t\mathbf{R}_t(\mathbf{R}_t'\mathbf{\Pi}_t\mathbf{R}_t)^{-1}$$

$$\mathbf{R}_{t+1} = \arg \min_{\mathbf{R}} f(\mathbf{L}_{t+1}, \mathbf{R}, \mathbf{\Pi}_t) = \mathbf{Y}'\mathbf{L}_{t+1}(\mathbf{L}_{t+1}'\mathbf{L}_{t+1})^{-1}$$

$$\mathbf{\Pi}_{t+1} = \arg \min_{\mathbf{\Pi}} f(\mathbf{L}_{t+1}, \mathbf{R}_{t+1}, \mathbf{\Pi}) = \text{diag}(p_1, \dots, p_N) \text{ where } p_i = D^{-1} \|(\mathbf{Y} - \mathbf{L}_{t+1}\mathbf{R}_{t+1}')_{:,i}\|_2^2$$

and we initialize by the PCA solution, that is:

$$\text{SpectralInit}(\mathbf{Y}) = \mathbf{U}\mathbf{\Sigma}\mathbf{V}' \approx (\mathbf{U}_{1:\hat{d}}\mathbf{\Sigma}_{1:\hat{d}}^{1/2})(\mathbf{\Sigma}_{1:\hat{d}}^{1/2}\mathbf{V}_{1:\hat{d}}') = (\mathbf{L}_0)(\mathbf{R}_0')$$

* more precisely called LR-ALPCA^H in our works

Related Methods

Robust PCA (RPCA)¹

Model Y as a sum of low-rank + sparse decomposition:

$$\arg \min_{\mathbf{X}, \mathbf{E}} \underbrace{\lambda \|\mathbf{X}\|_*}_{\text{low-rank}} + \underbrace{\|\mathbf{E}\|_{1,1}}_{\text{outliers}} \quad \text{s.t.} \quad \mathbf{Y} = \mathbf{X} + \mathbf{E}$$

Background Foreground Separation²



[1] Candès, Emmanuel J., et al. "Robust principal component analysis?." *Journal of the ACM (JACM)* 58.3 (2011): 1-37.

[2] <https://cs205tsqr.weebly.com/motivation.html>

Heteroscedastic Probabilistic PCA Technique (HePPCAT)¹

$\mathbf{y}$: measurement (known)
 $\mathbf{z}$: factor scores (unknown)
 $\boldsymbol{\epsilon}$: noise (unknown)
 i : sample index

Given G noise groups, a factor matrix $\mathbf{F}$, and scores $\mathbf{z}$, $\mathbf{y}$ is computed by:

$$\underbrace{\mathbf{y}_i \sim \mathcal{N}(\mathbf{0}, \overbrace{\mathbf{F}\mathbf{F}' + v_i\mathbf{I}}^{C_i})}_{\mathbf{y}_i = \mathbf{F}\mathbf{z}_i + \boldsymbol{\epsilon}_i} \quad \begin{array}{l} \boldsymbol{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, v_i\mathbf{I}) \\ \mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \end{array}$$

The negative log-likelihood based on this model is:

$$\mathcal{L}(\mathbf{F}, \mathbf{v}) = \frac{1}{2} \sum_{i=1}^N [-\log |C_i| - \mathbf{y}_i' C_i^{-1} \mathbf{y}_i]$$

and can be solved by expectation maximization (EM) on this likelihood.

[1] Hong, David, et al. "HePPCAT: Probabilistic PCA for data with heteroscedastic noise." *IEEE Transactions on Signal Processing* 69 (2021): 4819-4834.

Y : measurements
 U : subspace basis

Results

When the subspace is **known**

Subspace Affinity Error (SAE):

$$\text{SAE}(U, \hat{U}) = \frac{\|UU' - \hat{U}\hat{U}'\|_F}{\|UU'\|_F}$$

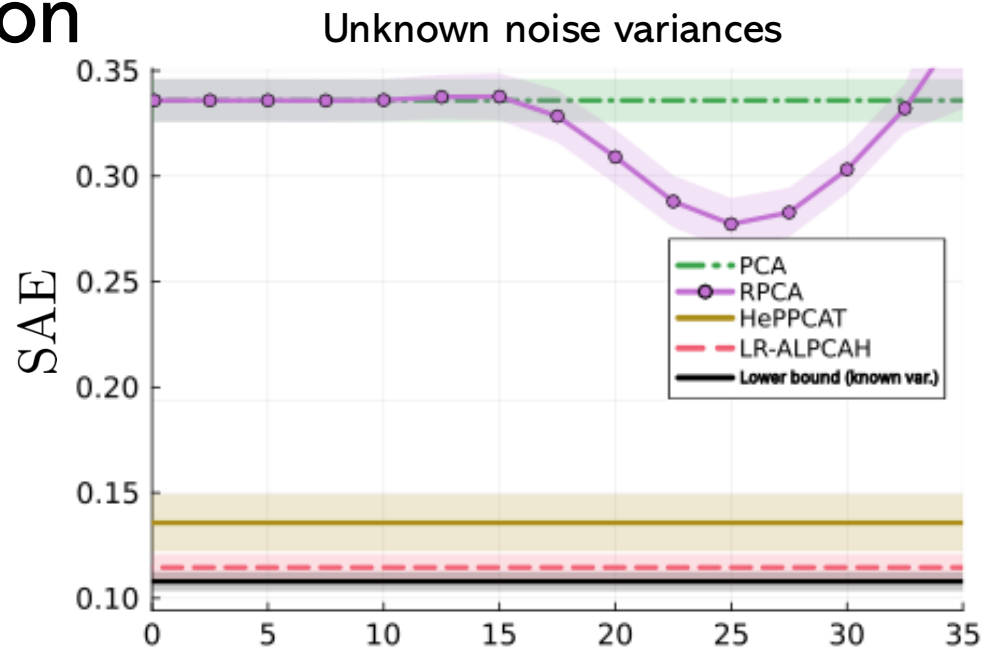
When the subspace is **unknown**

Normalized Root Mean Square Difference (NRMDS):

$$\text{NRMDS} = \frac{\|Y_{\text{test}} - U_{\text{train}}U'_{\text{train}}Y_{\text{test}}\|_F}{\|Y_{\text{test}}\|_F}$$

Synthetic Data Comparison

- We will compare SAE of all methods with true basis
- Only RPCA has a λ param.
- Many trials with different realizations of bases, noise, etc...
- **Uniform distribution** assumption for basis coordinates



λ (regularization for RPCA only)

Absolute subspace quality performance of ALPCA compared against other methods.

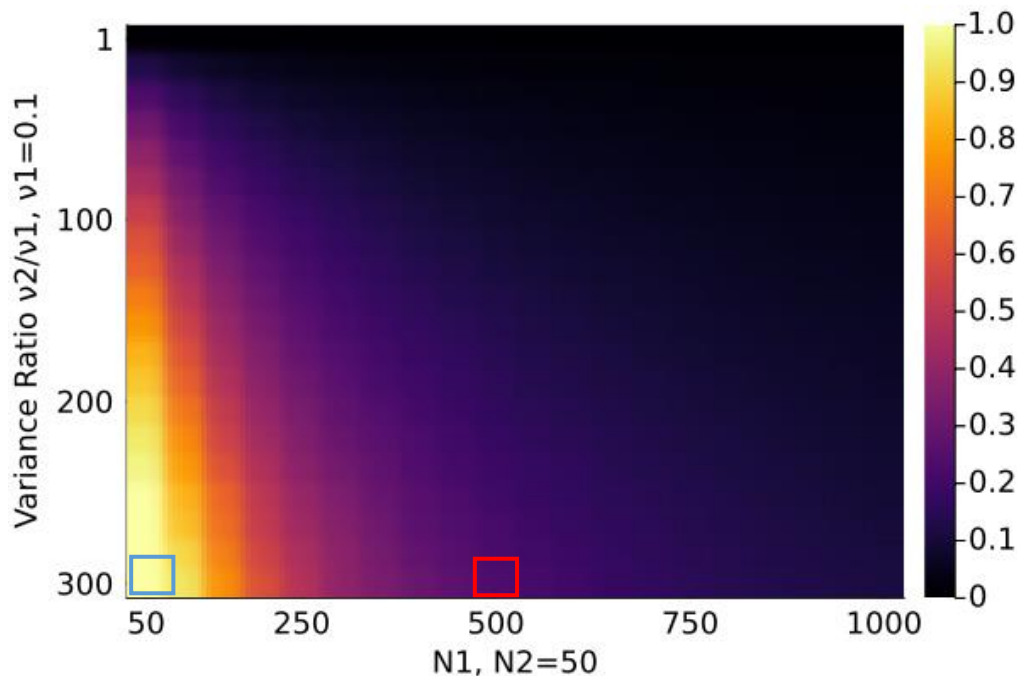
$$\text{SAE}(\mathbf{U}, \hat{\mathbf{U}}) = \|\mathbf{U}\mathbf{U}' - \hat{\mathbf{U}}\hat{\mathbf{U}}'\|_F / \|\mathbf{U}\mathbf{U}'\|_F$$

Good Data Experiment

- Heatmap pixels = SAE difference
- Two data groups
 - 1 = low noise “good” data
 - 2 = high noise “bad” data
- X-axis: vary # of “good” data
- Y-axis: vary noise in “bad” data

With a 50/50 split of “good/bad” data,
under high levels of noise,
ALPCAH >> PCA

Even when “good” data is 90% of total data,
there is still a good improvement over PCA!

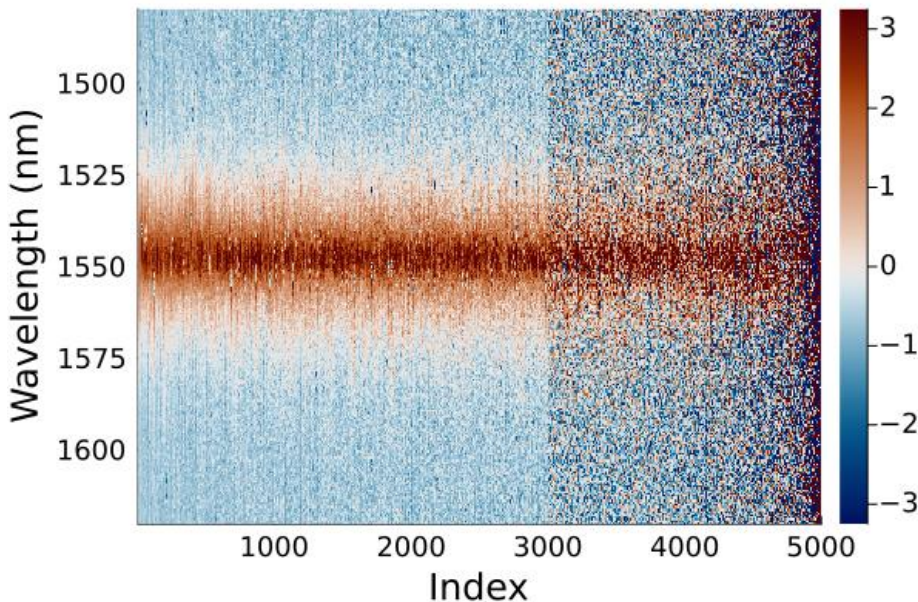


SAE(PCA) – SAE(ALPCAH) while good data amount varies.

$$\text{SAE}(\mathbf{U}, \hat{\mathbf{U}}) = \|\mathbf{U}\mathbf{U}' - \hat{\mathbf{U}}\hat{\mathbf{U}}'\|_F / \|\mathbf{U}\mathbf{U}'\|_F$$

Quasar Spectra Data^{1,2}

- Flux measurements across wavelengths for each quasar
- All data used is part of the same group
- Noise variances are estimated by SDSS team and we use this as reference values
- We use a high quality subset to compute “ground truth”

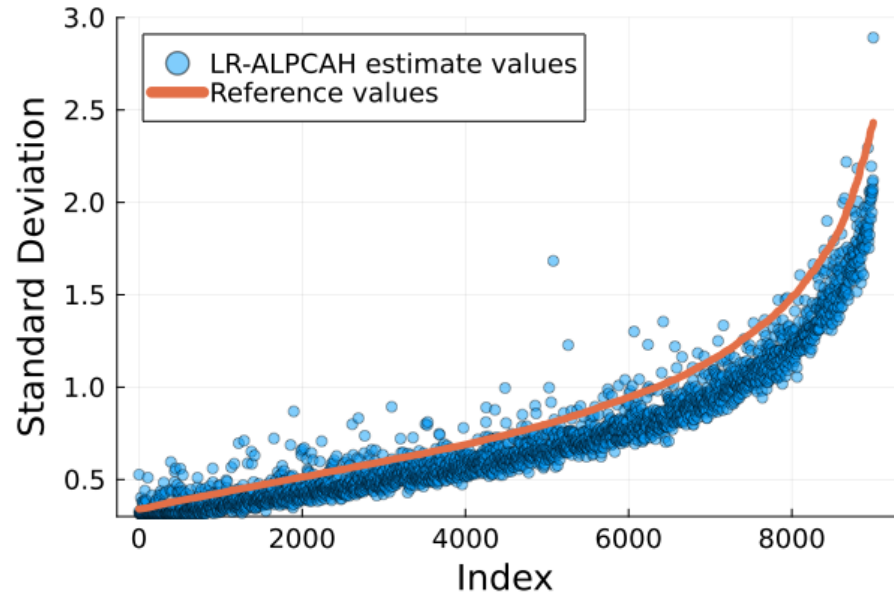


Sample data matrix of quasar flux measurements across wavelengths for each (column-wise) sample.

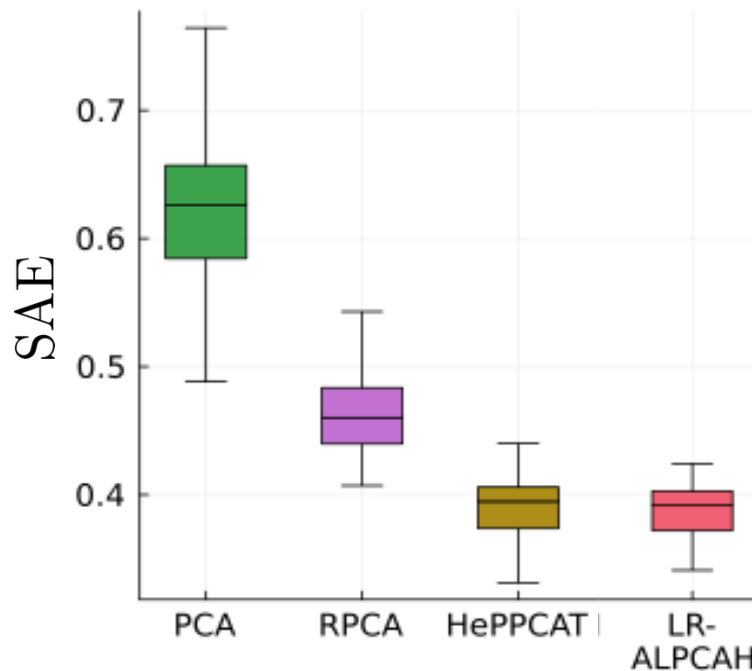
[1] Ahumada, Romina, et al. "The 16th data release of the Sloan Digital Sky Surveys: first release from the APOGEE-2 southern survey and full release of eBOSS spectra." *The Astrophysical Journal Supplement Series* 249.1 (2020): 3.

[2] Lyke, Brad W., et al. "The Sloan Digital Sky Survey Quasar Catalog: Sixteenth Data Release." *The Astrophysical Journal Supplement Series* 250.1 (2020): 8.

Quasar Data



Noise standard deviations estimates of LR-ALPCA compared to known astronomical reference values.



Subspace learning results showing SAE for ALPCA relative to other methods.

$$\text{SAE}(\mathbf{U}, \hat{\mathbf{U}}) = \|\mathbf{U}\mathbf{U}' - \hat{\mathbf{U}}\hat{\mathbf{U}}'\|_F / \|\mathbf{U}\mathbf{U}'\|_F$$

Quasar Spectra Data

	Time (ms)	Memory (MiB)	SAE (mean)
Homoscedastic Reference			
PCA	32.4	7.3	0.65
Heteroscedastic Methods			
RPCA	5091.8	7977.5	0.46
HePPCAT	1339.1	5731.6	0.39
ALPCAH	153.5	459.0	0.38

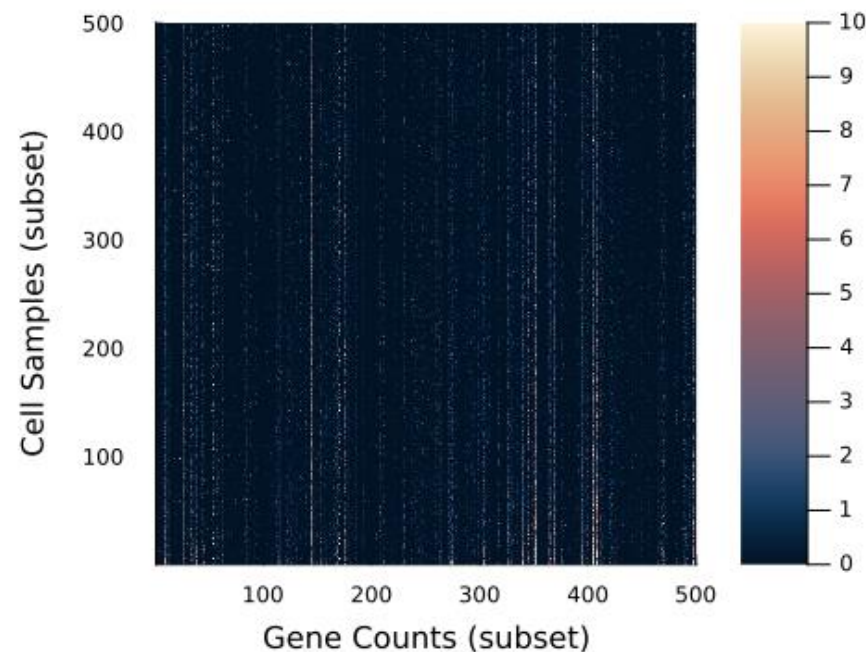
Subspace learning results on quasar flux data.

ALPCAH takes less time and memory for same performance!

$$\text{SAE}(\mathbf{U}, \hat{\mathbf{U}}) = \|\mathbf{U}\mathbf{U}' - \hat{\mathbf{U}}\hat{\mathbf{U}}'\|_F / \|\mathbf{U}\mathbf{U}'\|_F$$

Single Cell RNA Sequencing Data¹

- Quantifies the transcriptome of individual cells²
- Subspace learning useful for gene variational analysis and group/cluster analysis³
- Heteroscedasticity exists across both cells and genes⁴
- “Ground truth” not possible



Expression matrix based on cell samples (rows).

[1] Macosko, Evan Z., et al. "Highly parallel genome-wide expression profiling of individual cells using nanoliter droplets." *Cell* 161.5 (2015): 1202-1214.

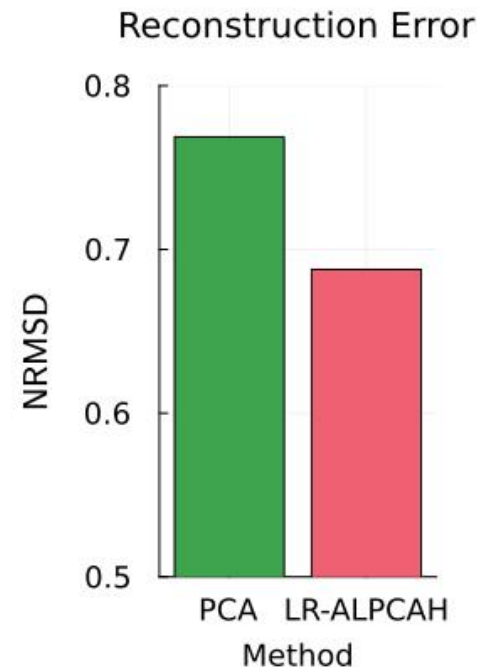
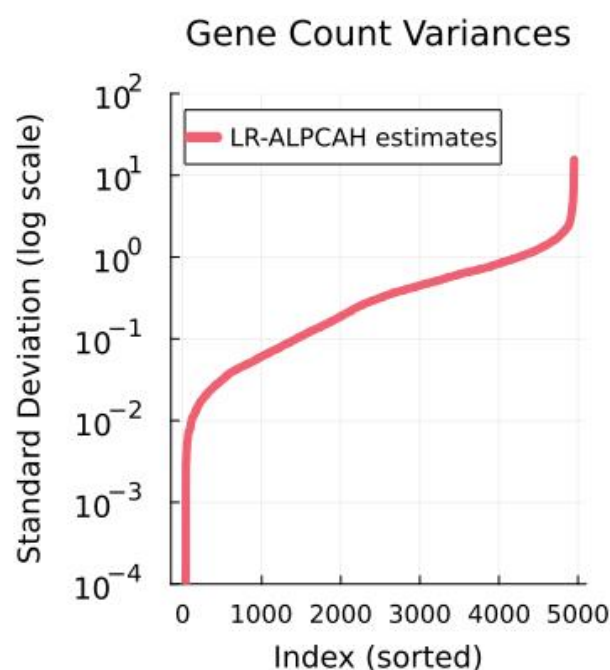
[2] Andrews, Tallulah S., and Martin Hemberg. "Identifying cell populations with scRNASeq." *Molecular aspects of medicine* 59 (2018): 114-122.

[3] Kharchenko, Peter V. "The triumphs and limitations of computational methods for scRNA-seq." *Nature methods* 18.7 (2021): 723-732.

[4] Hong, David, Yue Sheng, and Edgar Dobriban. "Selecting the number of components in PCA via random signflips." *arXiv preprint arXiv:2012.02985* (2020).

Single Cell RNA Sequencing Data

- Must use recon. error
- Split data into train and test
- Prior work showed $\hat{d} = 5$ to be a good estimate¹



LR-ALPCA results on scRNA sequencing data.

[1] Hong, David, Yue Sheng, and Edgar Dobriban. "Selecting the number of components in PCA via random signflips." *arXiv preprint arXiv:2012.02985* (2020).

$$\text{NRMSE} = \frac{\|\mathbf{Y}_{\text{test}} - \mathbf{U}_{\text{train}} \mathbf{U}_{\text{train}}' \mathbf{Y}_{\text{test}}\|_F}{\|\mathbf{Y}_{\text{test}}\|_F}$$

Summary

- Heteroscedasticity exists in various applications
- ALPCAHA jointly estimates subspace and data quality
- When properly modeled, noisier data does not have to degrade subspace estimation (can improve it actually*)
- Improved subspaces leads to more accurate analysis and conclusions

* not shown in this presentation

Dissertation

Chapter 3: ALPCA-H: Subspace Learning for Heteroscedastic Data

Chapter 4: ALPCA-HUS: Subspace Clustering for Heteroscedastic Data

Chapter 5: PET-TURTLE: Deep Unsupervised Support Vector Machines for Imbalanced Data Clusters

Chapter 6: Alzheimer's Disease Classification in Functional MRI With 4D Joint Temporal-Spatial Kernels in Novel 4D CNN Model

Part
2

Chapter 7: Behavior Score Prediction in Resting-State Functional MRI by Deep State Space Modeling^{1,2}

[1] J. Salazar Cavazos et. al., "Behavior Score Prediction in Resting-State Functional MRI for Alzheimer's Disease Spectrum by Deep State Space Modeling," in [The International Society for Magnetic Resonance in Imaging \(ISMRM\)](#), abs. xxxx, 2026

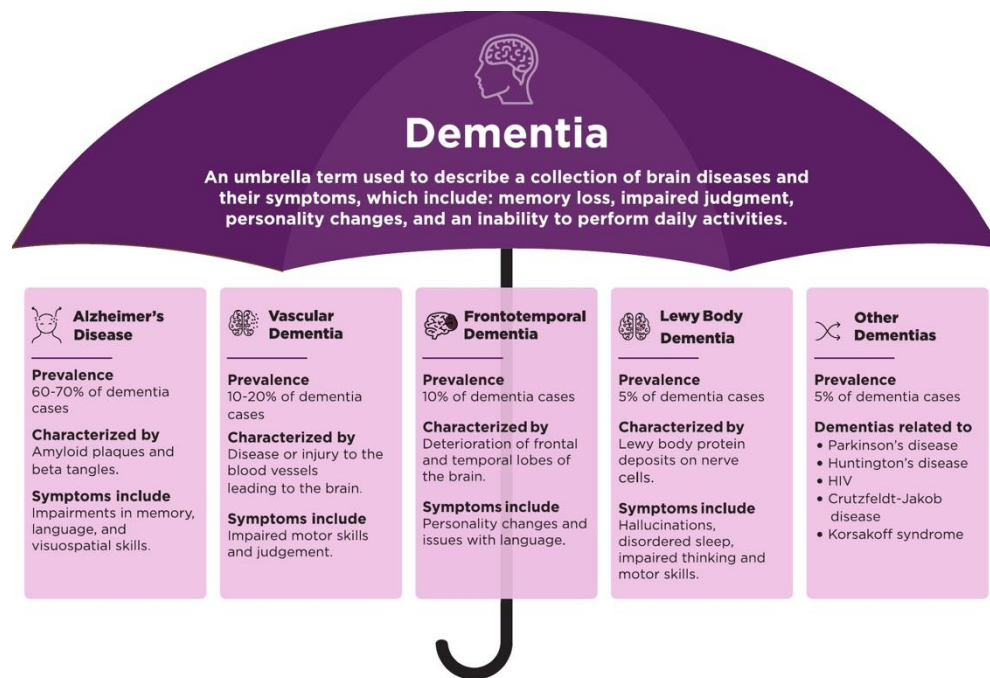
[2] J. Salazar Cavazos et. al., "Behavior Score Prediction in Resting-State Functional MRI by Deep State Space Modeling," [under review in IEEE Journal of Biomedical and Health Informatics](#), vol. xx, pp. xx-xx, 2026, doi: 10.48550/arXiv.2602.07131

Background

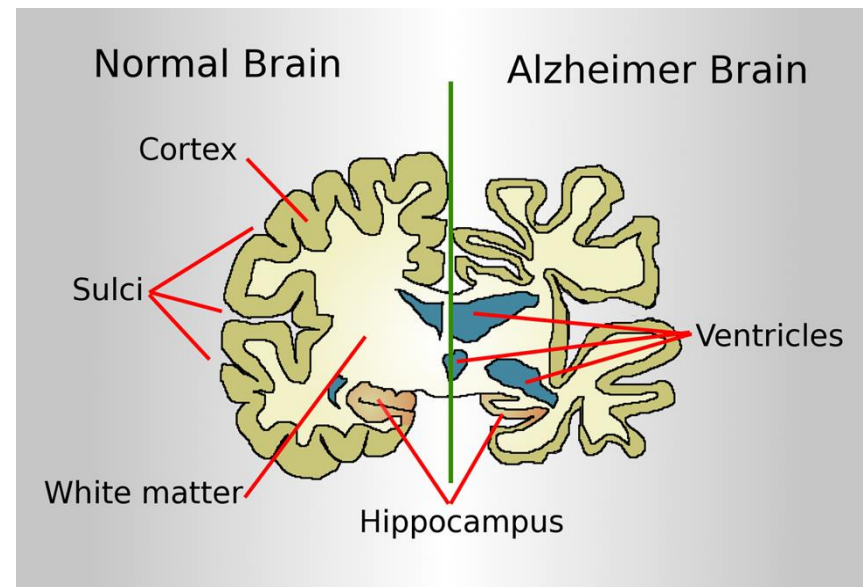
Functional MRI, Alzheimer's Disease, and Behavior Scores

Alzheimer's Disease (AD)

Dementia Umbrella¹



Effects of AD on structural brain²



[1] <https://www.summerfieldredlands.com/understanding-the-umbrella-of-dementia/>

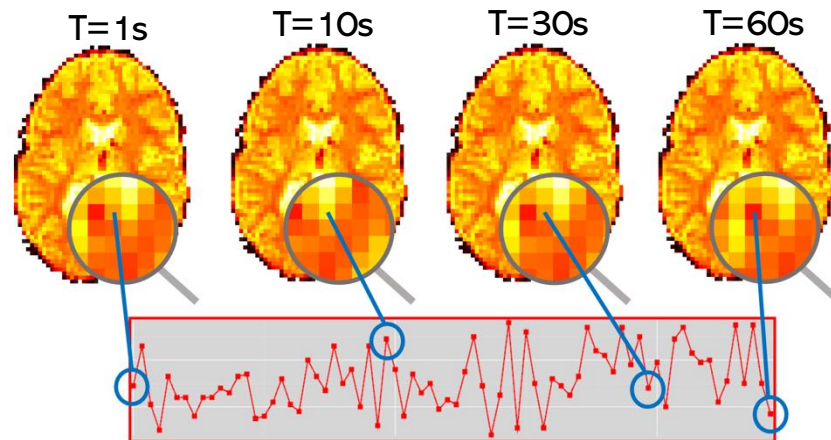
[2] <https://www.emf.ethz.ch/en/knowledge/topics/health/neurodegenerative-diseases/>

Functional MRI

MRI Scanner¹



Functional MRI²



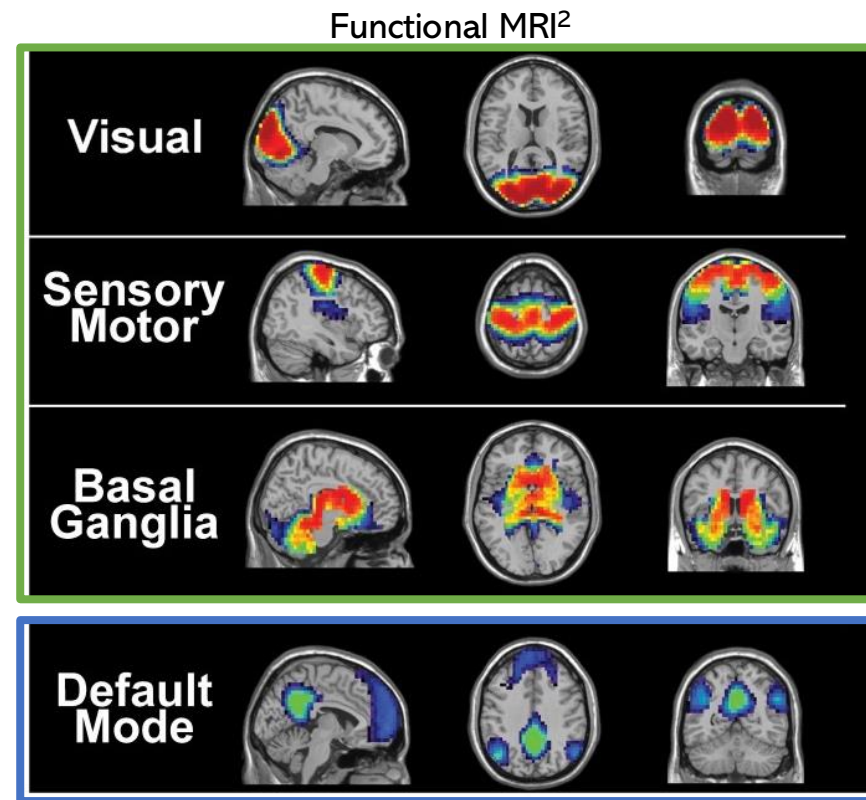
- 3D MRI volume acquired over time to form timeseries
- Active neurons require more oxygen
- BOLD data measures oxygenation changes in gray matter
- We use BOLD data as a **proxy** for neural activity

[1] <https://www.monash.edu/researchinfrastructure/mbi/facilities/human/3t-mri>

[2] <https://martynmcfarquhar.github.io/NCCN-IA-fMRIPreProcessing/1.fmri-structure.html>

Functional MRI

- Structural brain changes in AD happen **later**¹
- Functional changes happen several years **earlier**¹
- Two flavors of fMRI exist
 - Task-based
 - Resting-state



[1] Wierenga, Christina E., and Mark W. Bondi. "Use of functional magnetic resonance imaging in the early identification of Alzheimer's disease." *Neuropsychology review* 17.2 (2007): 127-143.

[2] Moussa, Malaak N., et al. "Consistency of network modules in resting-state FMRI connectome data." (2012): e44428.

MoCA & Diagnosis

- Easy and cheap early indicator for AD and mild impairment (MCI)
- MoCA achieves {accuracy: 89%, sensitivity: 90%, specificity: 87%} for MCI¹
- Doctors will use MoCA and other factors to determine diagnosis
- We look at MoCA and composite scores for {memory, language} categories

Montreal Cognitive Assessment (MoCA)

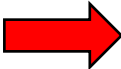
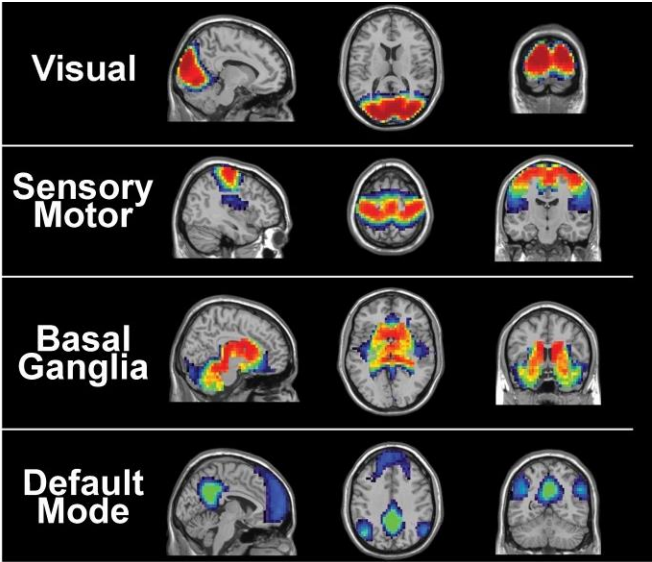
VISUOSPATIAL / EXECUTIVE		DRAW CLOCK (Ten past nine) (3 points)					POINTS
		☐ Contour ☐ Numbers ☐ Hands					
NAMING							___/3
MEMORY		Read list of words, subject must repeat them. Do 2 trials, even if 1st trial is successful. Do a recall after 5 minutes.					NO POINTS
		HAND	NYLON	PARK	CARROT	YELLOW	
1st TRIAL							
2nd TRIAL							

MoCA is a super set of visuospatial, memory, language, etc...

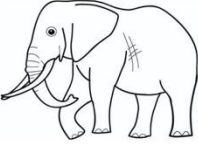
[1] Nasreddine, Ziad S., et al. "The Montreal Cognitive Assessment, MoCA: a brief screening tool for mild cognitive impairment." *Journal of the American Geriatrics Society* 53.4 (2005): 695-699.

Behavior Score Prediction

Functional MRI¹



Montreal Cognitive Assessment² (MoCA)

VISUOSPATIAL / EXECUTIVE		Copy chair	Draw CLOCK (Ten past nine) (3 points)	POINTS				
[]		[]	[] [] [] Contour Numbers Hands	___/5				
NAMING								
 []  []  []								
___/3								
MEMORY		HAND NYLON PARK CARROT YELLOW						
Read list of words, subject must repeat them. Do 2 trials, even if 1st trial is successful. Do a recall after 5 minutes.		1st TRIAL						NO POINTS
		2nd TRIAL						

Given Functional MRI data, predict behavior scores!

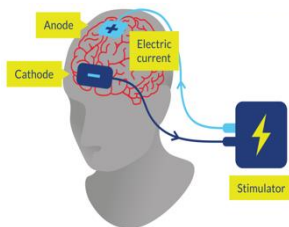
[1] Moussa, Malaak N., et al. "Consistency of network modules in resting-state fMRI connectome data." (2012): e44428.
[2] Nasreddine, Ziad S., et al. "The Montreal Cognitive Assessment, MoCA: a brief screening tool for mild cognitive impairment." *Journal of the American Geriatrics Society* 53.4 (2005): 695-699.

Motivation

- Functional MRI will not replace MoCA
- Understand brain-behavior relationship
- Identify impaired brain regions
- Could improve diagnosis/monitoring/intervention for AD

ABSTRACT ONLY (NO FULL PAPER YET):

Synergistic effects of cognitive training and high-definition transcranial direct current stimulation across the dementia spectrum¹



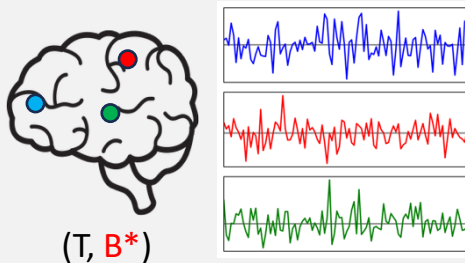
[1] Only targets the
left ventrolateral
prefrontal cortex!

[1] Hampstead, Benjamin M., et al. "Synergistic effects of cognitive training and high-definition transcranial direct current stimulation across the dementia spectrum." *Brain Stimulation: Basic, Translational, and Clinical Research in Neuromodulation* 18.1 (2025): 350.

Related Methods

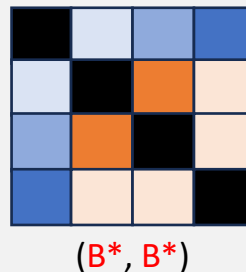
Functional MRI Data Views

Timeseries Data



- $T = 570$ time samples
- ~ 10 minute scan
- Temporal dynamics retained
- Many approaches to extract “features”

Functional Connectivity Data



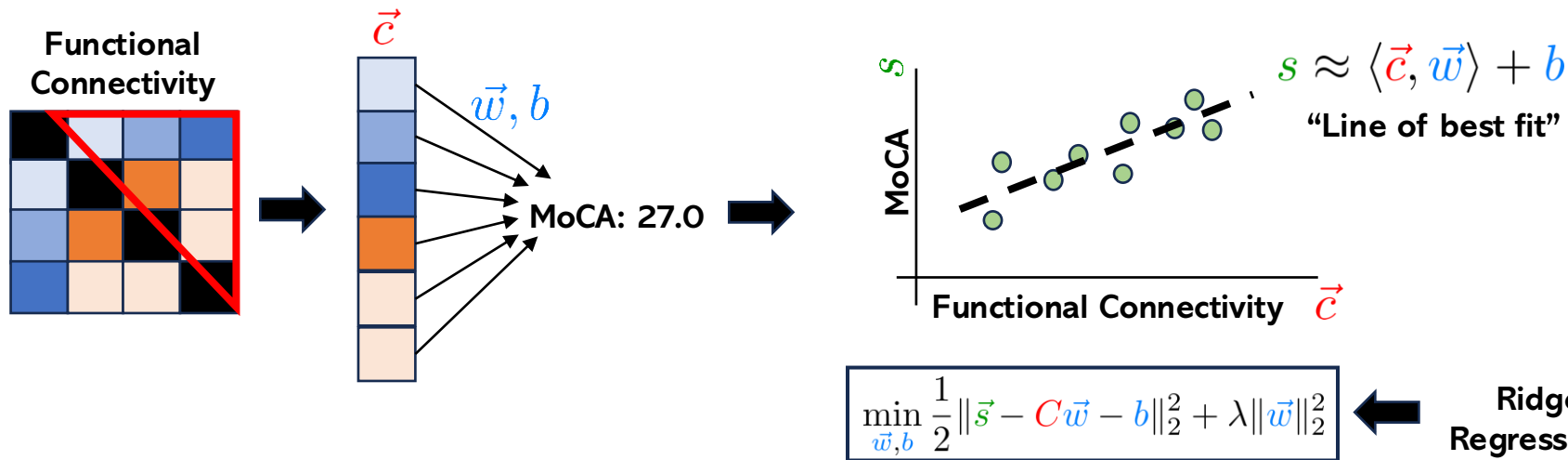
- $B^* = 272$ brain regions
- Correlation matrix of regions' timeseries data
- Highly popular in the field to study networks

* In this work, we use/augment the Power¹ atlas to extract 272 regions

[1] Power, Jonathan D., et al. "Functional network organization of the human brain." *Neuron* 72.4 (2011): 665-678.

Related Method 1 – FCM + KRR*

$\vec{c}$: connectivity for one sub.
 C : set of $\vec{c}$, $\forall$ subjects
 $\vec{w}, b$: learnable weights
 s : MoCA scores

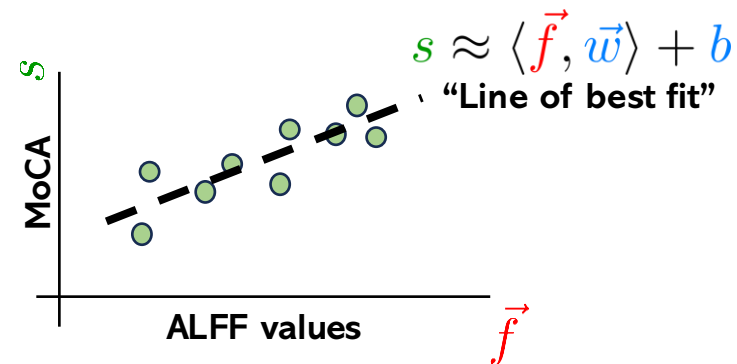
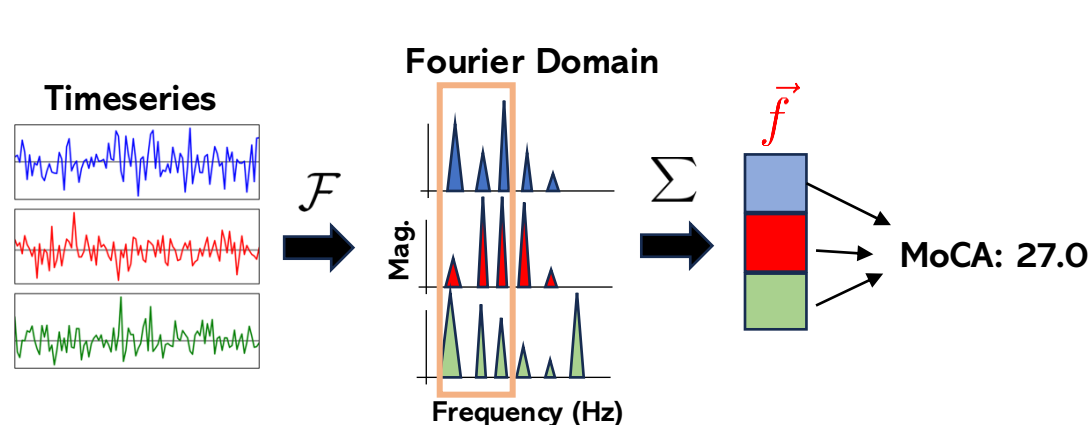


* In practice, kernel ridge regression is used to handle possible non-linearity

Related Method 2 – ALFF¹ + KRR^{*}

ALFF = Amplitude of Low Frequency Fluctuations

$\vec{f}$: ALFF for one sub.
 F : set of $\vec{f}$, $\forall$ subjects
 $\vec{w}, b$: learnable weights
 s : MoCA scores



$$\min_{\vec{w}, b} \frac{1}{2} \|\vec{s} - F\vec{w} - b\|_2^2 + \lambda \|\vec{w}\|_2^2$$

Ridge
Regression^{*}

^{*} In practice, kernel ridge regression is used to handle possible non-linearity

[1] Yang, Hong, et al. "Amplitude of low frequency fluctuation within visual areas revealed by resting-state functional MRI." *Neuroimage* 36.1 (2007): 144-152.

A Primer on State Space Models

State Space Models (SSMs)

Given input signal $u(t)$, generate output sequence $y(t)$ using state $x(t)$

$$\left. \begin{aligned} x'(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \end{aligned} \right\} \text{This is an LTI system}$$

A = State Matrix

B = Input Matrix

C = Output Matrix

D = Feedthrough Matrix

Toy Example

Consider a mass attached to the wall with a spring.

Let m = mass, k = spring constant, and b = friction constant

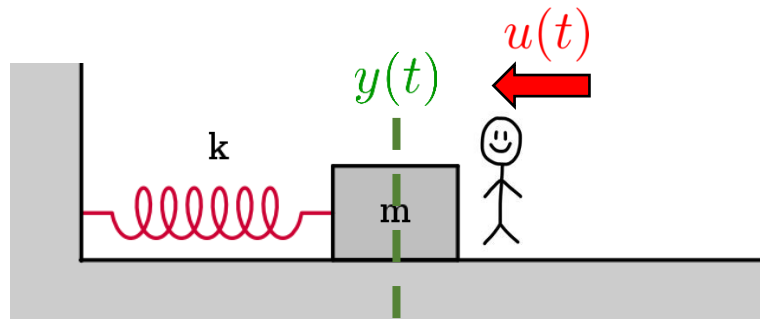
$u(t)$ is force applied to spring and $y(t)$ is position of object

$$my''(t) = u(t) - by'(t) - ky(t)$$



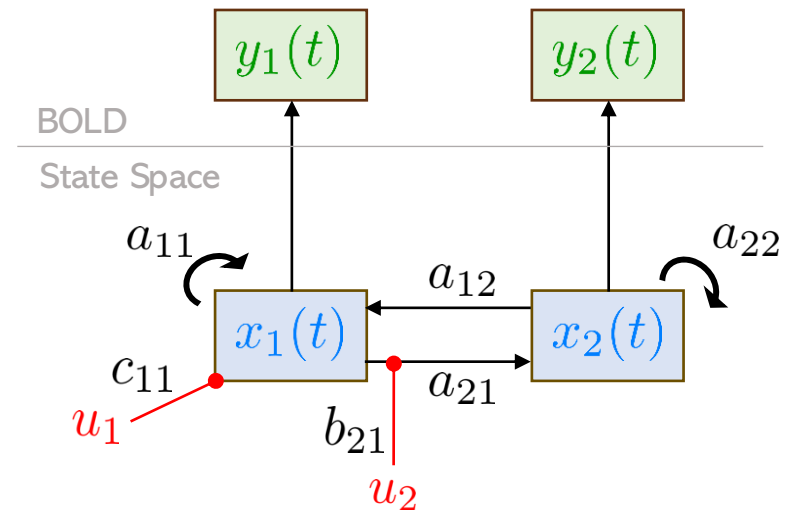
$$A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}, B = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

There are no Newtonian/Maxwell equations for fMRI!



Dynamic Causal Modeling¹

- DCM uses this state space idea to model BOLD timeseries as function of state variables
- Originally used for auditory word processing
- **Psychiatric Applications²**: schizophrenia analysis and to study ketamine effects on depression subjects
- Limitations:
 - Computationally challenging beyond 10 nodes
 - Linear approximations (LTI assumptions)



$$x'_1 = a_{11}x_1 + a_{12}x_2 + c_{11}u_1$$

$$x'_2 = a_{21}x_1 + a_{22}x_2 + b_{21}u_2$$

A : intrinsic (fixed) connectivity

B : modulatory effect

C : direct (driving) effect

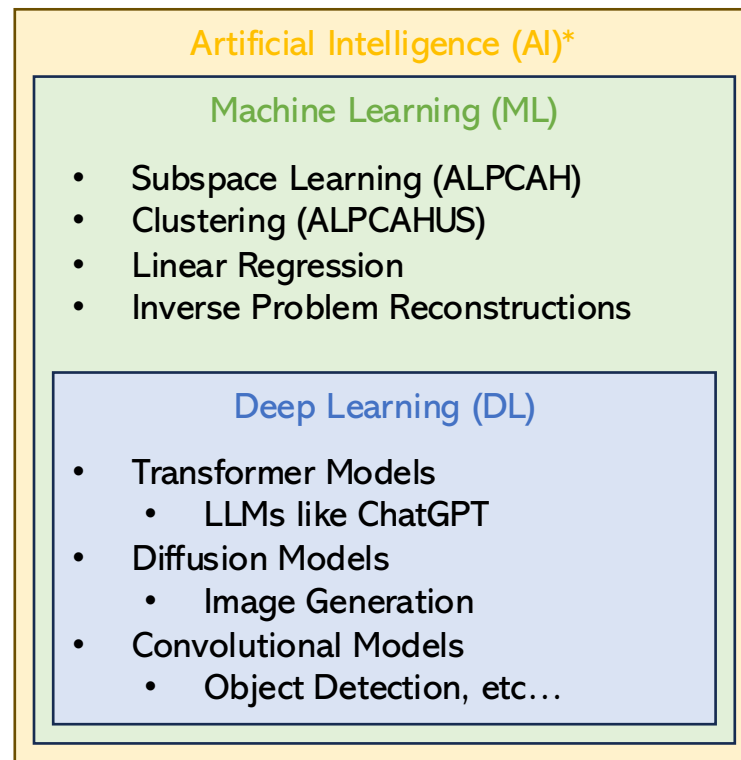
[1] Friston, Karl J., Lee Harrison, and Will Penny. "Dynamic causal modelling." *Neuroimage* 19.4 (2003): 1273-1302.

[2] Heinzle, Jakob, and Klaas E. Stephan. "Dynamic causal modeling and its application to psychiatric disorders." *Computational psychiatry*. Academic Press, 2018. 117-144.

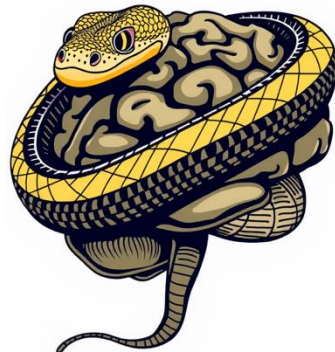
Deep Learning (DL)

- DCM is not feasible for our 272 regions
- Deep learning is a viable alternative methodology
- Two reasons for recent popularity:
 - GPU >> CPU
 - Non-linear modeling

Can we merge state space modeling and deep learning to handle limitations with dynamic causal modeling?



* AI means using math modeling and data to solve problems (my perspective)



Method

NeuroMamba

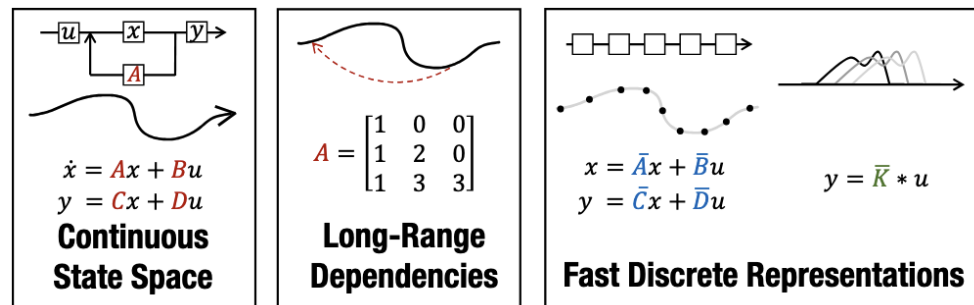
Deep State Space Modeling

FLUX.1 generated image with prompt “an illustration of a mamba snake wrapped around a human brain”

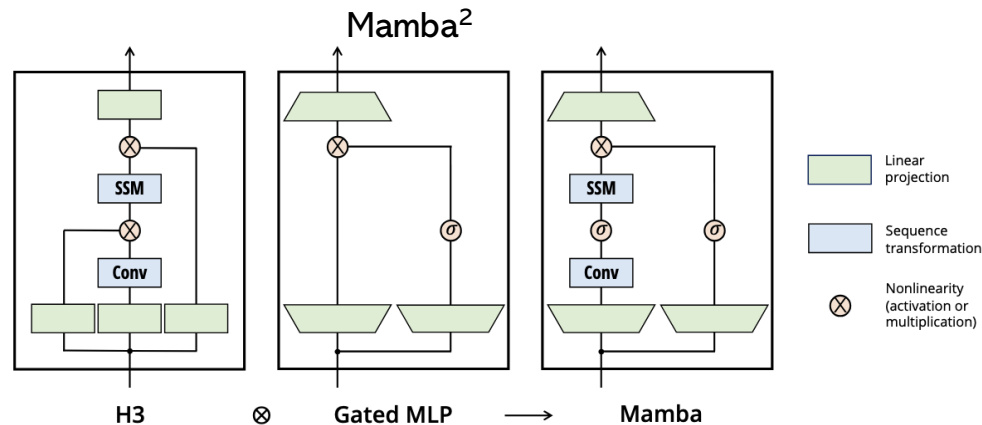
Deep SSMs: S4 & Mamba

- S4 creates a discrete representation of a continuous state space
- Smart initialization to avoid unstable system
- LTI system allows for fast computations

Structured State Space Sequence Model¹ (S4)



- Improves upon S4 by integrating nonlinearity in "Mamba" block itself
- No longer time invariant system for more complex modeling
- Uses parallel scan "tricks" for fast computation



[1] Gu, Albert, Karan Goel, and Christopher Ré. "Efficiently modeling long sequences with structured state spaces." *arXiv preprint arXiv:2111.00396* (2021).

[2] Gu, Albert, and Tri Dao. "Mamba: Linear-time sequence modeling with selective state spaces." *First conference on language modeling*. 2024.

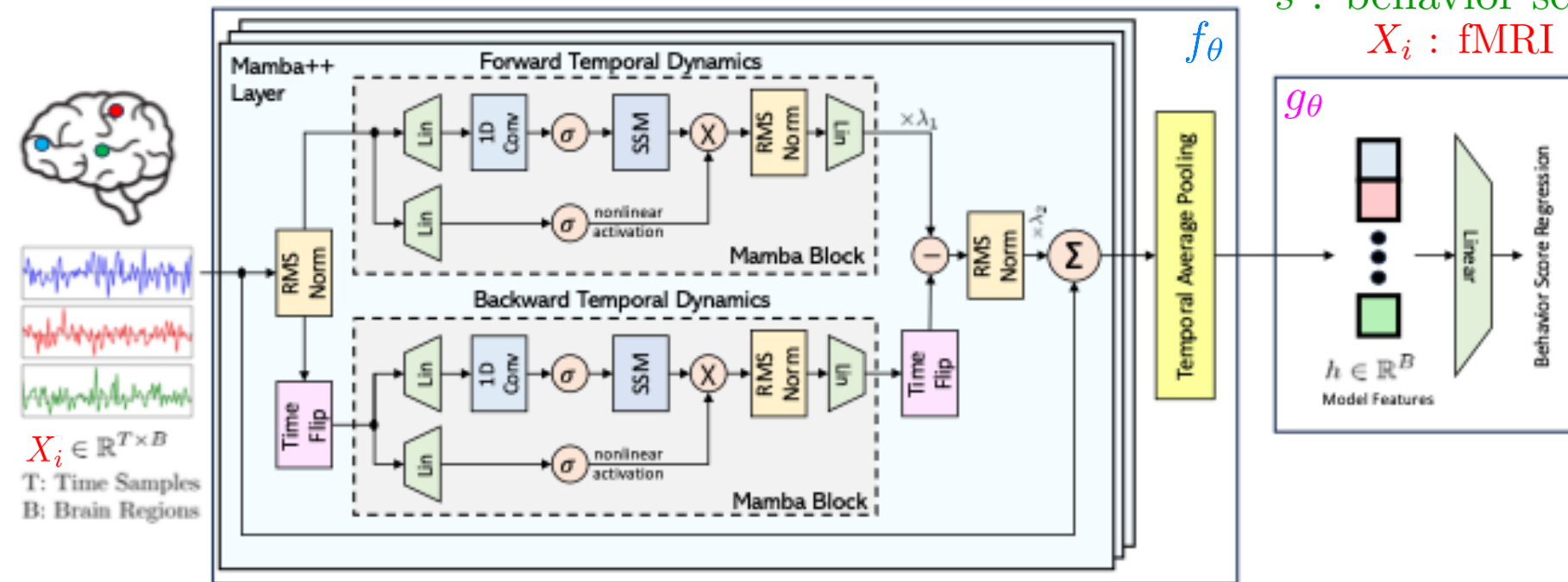
NeuroMamba

f_θ : NeuroMamba backbone

g_θ : readout function

$\vec{s}$: behavior scores

X_i : fMRI data



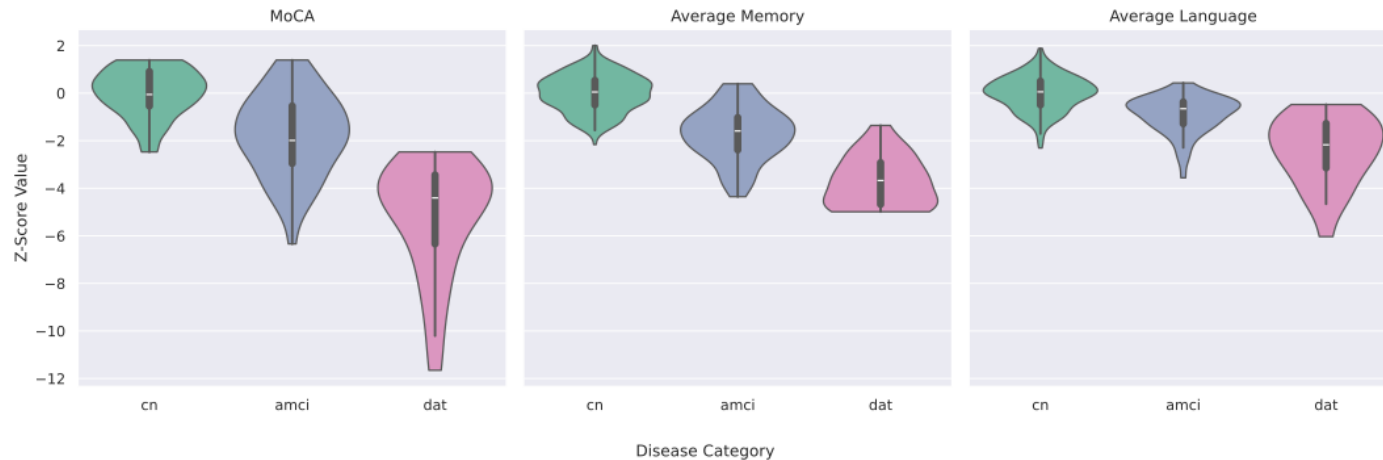
$$\min_{\theta} \sum_i \underbrace{\|\vec{s}_i - g_\theta(f_\theta(X_i))\|_2^2}_{\text{data fidelity}} + \underbrace{\lambda \|f_\theta(X_i)\|_1}_{\text{sparse brain regions}}$$

Resting-State Functional MRI Results

MADRC Data

Category	Total Subjects	Race (W/B/A)	Sex (M/F)	Average MoCA
CN	203	169/31/3	132/71	27.1
aMCI	53	44/8/1	16/37	23.5
DAT	25	24/0/1	12/13	16.3

Distribution of subjects in the MADRC dataset, stratified by disease category.



Violin plots illustrating the distribution of behavioral scores in z-score space across different disease categories.

Correlation Values

Functional Connectivity {	Method	MoCA	Memory	Language
	FCM + KRR	0.07	0.08	0.10

Aligns with existing MoCA literature: 0.07¹ [AD] and 0.15² [LLD]

Pearson correlation coefficients and corresponding p-values by score category. Asterisks denote statistical significance as follows: * = $p < 0.1$, ** = $p < 0.01$, *** = $p < 0.001$

[1] Karker, Michelle. *Predictive Analysis and Deep Learning of Functional MRI in Alzheimer's Disease*. Diss. 2022.

[2] Liu, Chaomeng, et al. "Individualized prediction of cognitive test scores from functional brain connectome in patients with first-episode late-life depression." *Journal of Affective Disorders* 352 (2024): 32-42.

[3] Yang, Liu, et al. "Gradual disturbances of the amplitude of low-frequency fluctuations (ALFF) and fractional ALFF in Alzheimer spectrum." *Frontiers in neuroscience* 12 (2018): 975.

Correlation Values

	Method	MoCA	Memory	Language
Functional Connectivity	FCM + KRR	0.07	0.08	0.10
	I-ICA + KRR	0.14*	0.09	0.10*
Model-based Timeseries	G-ICA + KRR	0.18**	0.12*	0.16**
	ALFF + KRR	0.20***	0.13*	0.18**

Aligns with existing MoCA literature: 0.07¹ [AD] and 0.15² [LLD]

Aligns with existing MMSE literature: 0.21³ [AD]

Pearson correlation coefficients and corresponding p-values by score category. Asterisks denote statistical significance as follows: * = $p < 0.1$, ** = $p < 0.01$, *** = $p < 0.001$

[1] Karker, Michelle. *Predictive Analysis and Deep Learning of Functional MRI in Alzheimer's Disease*. Diss. 2022.

[2] Liu, Chaomeng, et al. "Individualized prediction of cognitive test scores from functional brain connectome in patients with first-episode late-life depression." *Journal of Affective Disorders* 352 (2024): 32-42.

[3] Yang, Liu, et al. "Gradual disturbances of the amplitude of low-frequency fluctuations (ALFF) and fractional ALFF in Alzheimer spectrum." *Frontiers in neuroscience* 12 (2018): 975.

Correlation Values

	Method	MoCA	Memory	Language
Functional Connectivity	FCM + KRR	0.07	0.08	0.10
	I-ICA + KRR	0.14*	0.09	0.10*
Model-based Timeseries	G-ICA + KRR	0.18**	0.12*	0.16**
	ALFF + KRR	0.20***	0.13*	0.18**
Data-driven Timeseries	TCN (CNN)	0.26***	0.16**	0.17**
	BiLSTM (RNN)	0.19***	0.19***	0.18**
	PatchTST (Trans.)	0.28***	0.17**	0.19***
	NeuroMamba	0.36***	0.24***	0.25***

Aligns with existing MoCA literature: 0.07¹ [AD] and 0.15² [LLD]

Aligns with existing MMSE literature: 0.21³ [AD]

Pearson correlation coefficients and corresponding p-values by score category. Asterisks denote statistical significance as follows: * = $p < 0.1$, ** = $p < 0.01$, *** = $p < 0.001$

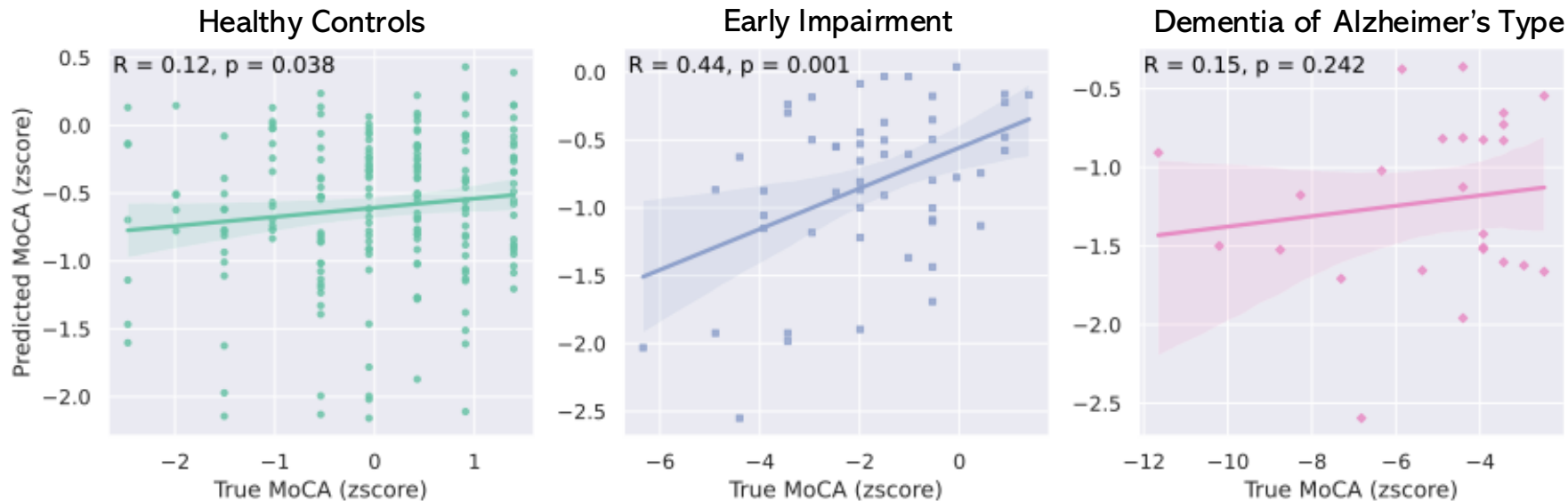
data-driven timeseries methods >> model-based timeseries methods >> functional connectivity

[1] Karker, Michelle. *Predictive Analysis and Deep Learning of Functional MRI in Alzheimer's Disease*. Diss. 2022.

[2] Liu, Chaomeng, et al. "Individualized prediction of cognitive test scores from functional brain connectome in patients with first-episode late-life depression." *Journal of Affective Disorders* 352 (2024): 32-42.

[3] Yang, Liu, et al. "Gradual disturbances of the amplitude of low-frequency fluctuations (ALFF) and fractional ALFF in Alzheimer spectrum." *Frontiers in neuroscience* 12 (2018): 975.

Scatter Plots



Correlation scatter plots displaying the relationship between predicted NeuroMamba scores and true behavioral metrics (MoCA), across the Alzheimer's disease spectrum (columns), presented in z-score normalized space.

highest correlation among amnesic MCI subjects ($R=0.44$)

Impacted Brain Regions

PFI ¹ (RMSE)	Power ROI	MNI (x,y,z)	Nominal System	Lobe/Area	Talairach Daemon Label
MoCA					
1.84	77	(-13, -40, 1)	Default Mode	Limbic Cortex	Parahippocampal Gyrus
0.39	170	(6, -81, 6)	Visual	Occipital Lobe	Cuneus
0.21	177	(-53, -49, 43)	Fronto-parietal Task Control	Parietal Lobe	Inferior Parietal Lobule
0.07	212	(-11, 26, 25)	Salience	Limbic Cortex	Anterior Cingulate
0.05	93	(15, -63, 26)	Default Mode	Occipital Lobe	Precuneus

Nominal Purpose: memory encoding, retrieval, spatial navigation, and contextual processing

AD: atrophy, decline in episodic memory, early deposition of amyloid-beta plaque and tau proteins, disrupted DMN

Nominal Purpose: higher-order visual processing,

AD: deficits in visual processing, spatial navigation, and attentional functions

Nominal Purpose: speech, language, number processing, working memory

AD: atrophy of bank thickness, abnormal functional activity with salience/sensory/executive networks

Nominal Purpose: emotion regulation, impulse control, motivation, self-directed behavior

AD: high agitation/irritability/anxiety due to atrophy and reduced functional connectivity

Nominal Purpose: daydreaming, self-reflection, mental imagery

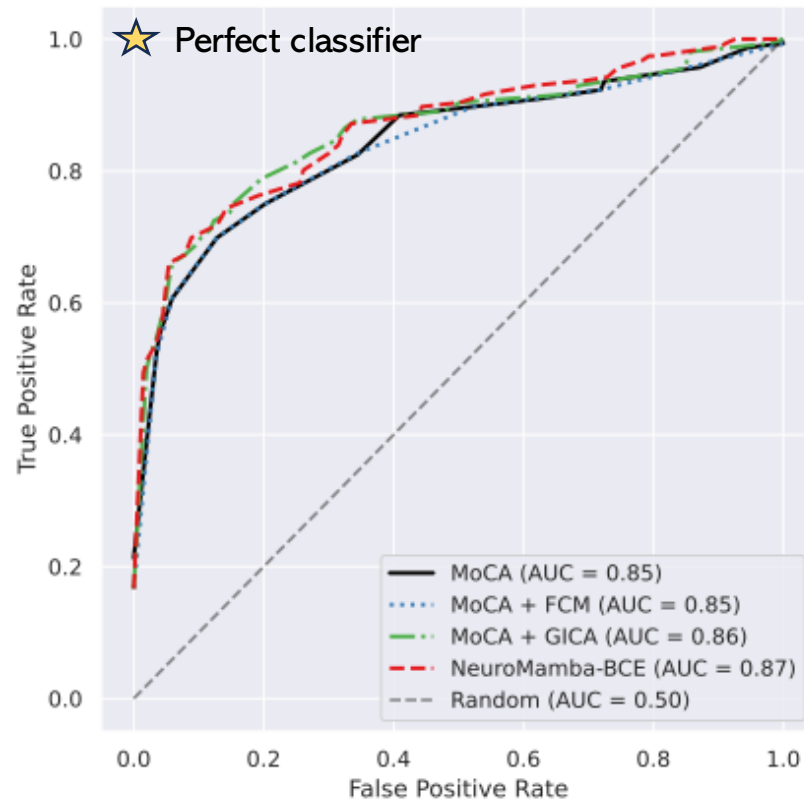
AD: reduced metabolism and atrophy

[1] Altmann, André, et al. "Permutation importance: a corrected feature importance measure." *Bioinformatics* 26.10 (2010): 1340-1347.

Binary Diagnosis using MoCA + fMRI

- Does fMRI provide complimentary information for better diagnosis? Or is MoCA alone enough?
- Binary setting of simple healthy and non-healthy
- All methods use MoCA + fMRI features
- Feature-based methods use logistic regression
- NeuroMamba trained in a cross entropy setup

No significant AUC improvement when adding any kind of fMRI features!



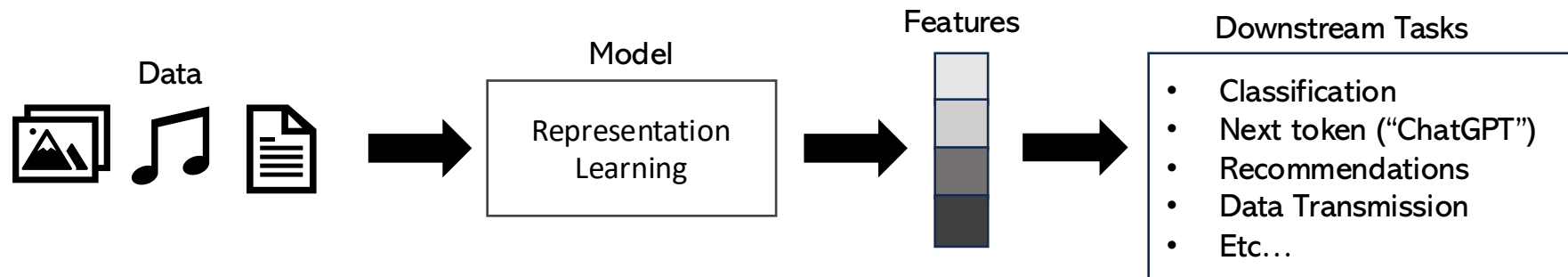
Receiver operating characteristic (ROC) curve with area under curve (AUC) values for diagnosis between cognitively normal and non-normal subjects using MoCA score in conjunction with rs-fMRI features.

Summary

- Resting-state fMRI data has moderate correlation to behavior scores ($R=0.36$)
- Data-driven timeseries methods >> functional connectivity
- Resting-state fMRI does not seem to provide complimentary info for diagnosis
- Implicated brain regions could be used in intervention strategies such as stimulation

Wrapping things up...

Representation Learning



- Part 1 covered explicit modeling (subspace learning)
- Part 2 covered implicit modeling (deep learning)
- Both are flavors of representation learning

Future Work

- Chapter 3: ALPCAH: Subspace Learning for Heteroscedastic Data
 - Doubly heteroscedastic modeling
 - Poisson noise distributions
 - Tensor modeling (as opposed to matrices)
 - Heteroscedasticity in DL applications
- Chapter 7: Behavior Score Prediction in Resting-State Functional MRI by Deep SSMs
 - Task-based functional MRI & behavioral scores
 - Functional MRI & blood tau protein biomarker relationship
 - Longitudinal study on disease progression (beyond baseline scans)

Acknowledgements



National Institutes
of Health

Scott; Grant R21 AG082204
MADRC; P30 AG07293



National
Science
Foundation

Laura; CAREER Grant CCF-1845076
Laura; Grant CCF-2331590



UM; Rackham Merit Fellowship

Acknowledgements



Acknowledgements

My family*



My second family*



* wedding pictures

Acknowledgements



Jeff Fessler

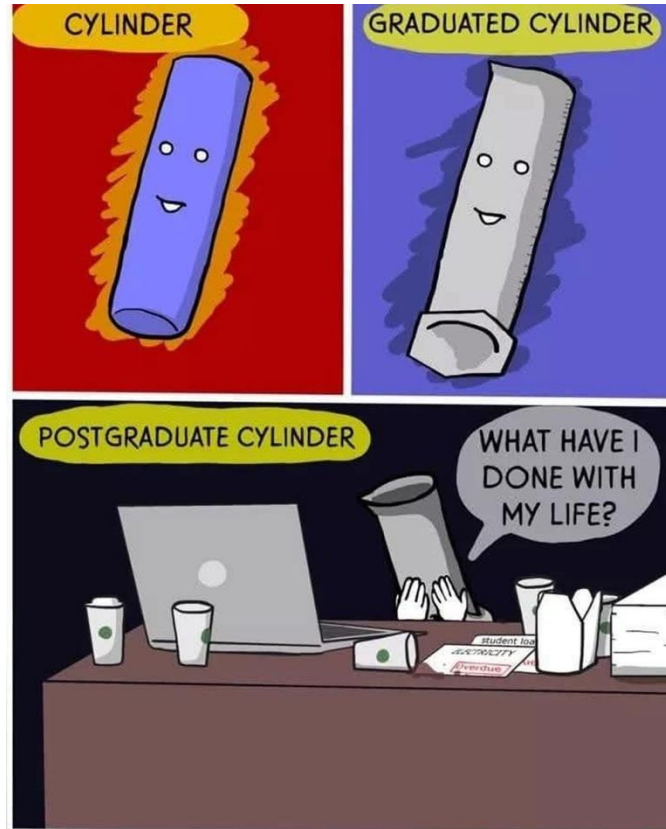


Laura Balzano



Scott Peltier

Thanks for listening! Questions?



unknown source